SAT Math Mastery · Session 8

Data Models, Angles and Triangles

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 45 minutes), and a short error log.
Topics	scatterplots and lines of best fit: predicted versus actual, slope and intercept in context, linear versus exponential models · parallel lines and transversals · angle sums and exterior angles · isosceles triangles and regular polygons · similar triangles and proportional segments · triangle inequality, midsegments and angle bisectors · right-triangle trigonometry and special right triangles · Law of Cosines
Name / Date	_____ _____

PART A · IN CLASS — 37 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. MEDIUM · DESMOS

A household water-conservation study recorded, for several summer days, the daily high temperature x (in degrees Celsius above 25 °C) and the household's water use y (in liters). A line of best fit for the data is given by $y = 12x+20$. On one recorded day, $x = 10$ and the household's actual water use was 155 liters. What is the positive difference between the actual water use on that day and the water use predicted by the line of best fit?

1. Type the model `y=12x+20` and the recorded day `(10,155)`. The point sits above the line.
2. The predicted value at $x = 10$: Type `12*10+20`: it returns 140 liters.
3. The positive difference is actual minus predicted: Type `155-(12*10+20)`: it returns 15.

Answer: 15

2. MEDIUM · DESMOS

A scatterplot shows the relationship between x, the distance in kilometers from a city's central waterfront, and y, the number of visitors counted per hour at a beach access point. The line of best fit for the data passes through the points $(2, 340)$ and $(10, 60)$. What is the slope of this line of best fit?

(A) 35

(B) −35 ✓

(C) −28

(D) 28

1. Type the two points `(2,340)` and `(10,60)`.
2. Slope = rise over run: Type `(60-340)/(10-2)`: it returns −35.
3. Check by drawing the line through both points: type `y=-35(x-2)+340`. At $x = 2$ Type `-35*(2-2)+340`: it returns 340 and at $x = 10$ Type `-35*(10-2)+340`: it returns 60, so the line passes through both points.
4. The slope is −35 (the count falls as distance grows), option (B).

Answer: (B)

3. Medium · Desmos

A line of best fit models y, the installed solar capacity (in megawatts) at a pilot site, in terms of x, the number of years since 2015: $y = 45.2x + 12.5$. Which of the following is closest to the capacity predicted by this model at $x = 8.4$?

(A) 380.5 (B) 388.7

(C) 392.2 ✓ (D) 401.0

1. Type the model `y=45.2x+12.5` (letting x be the years since 2015).
2. Evaluate it at $x = 8.4$: Type `45.2*8.4+12.5`: it returns 392.18.
3. Round to one decimal place: Type `round(45.2*8.4+12.5,1)`: it returns 392.2.
4. That is option (C).

Answer: (C)

4. Medium · No calculator needed

A water-treatment plant's storage tank begins a shift completely full. The tank is drained at a constant rate throughout the shift to supply a neighborhood. What type of model best describes the volume of water remaining in the tank at any time during the shift?

(A) A linear model with a negative slope ✓ (B) An exponential growth model

(C) An exponential decay model (D) A linear model with a positive slope

1. Open a table and enter $x_1 = 0, 1, 2, 3$ and $y_1 = 1000, 915, 830, 745$. (a tank draining 85 liters per hour). Type `y_1~mx_1+b`: Desmos returns $m = -85$, $b = 1000$, $r^2 = 1$, a perfect line with negative slope.
2. Compare a decay that removes 10 percent each hour: Open a table and enter $x_1 = 0, 1, 2, 3$ and $y_1 = 1000.0, 900.0, 810.0, 729.0$. Type `y_1~mx_1+b`: Desmos returns $r^2 = 0.9978$ and the points curve away from the line (the drops 100, 90, 81 are not constant).
3. A constant rate of drainage gives constant drops, so the model is linear with a negative slope, option (A). (Verbal reasoning step: the table and regression are only a model of the idea.)

Answer: (A)

5. Medium · No calculator needed

A scatterplot shows exam-preparation hours x and practice-exam score y for 10 students, with a line of best fit shown. For the student with the highest practice-exam score, the actual score was 96, and the score predicted by the line of best fit was 89. Which of the following is closest to the difference between the actual score and the predicted score for this student?

(A) 4 (B) 1

(C) 10 **(D) 7 ✓**

1. The difference is actual minus predicted: Type `96-89`: it returns 7.
2. Check by adding back: Type `89+7`: it returns 96, the actual score.
3. The point is 7 above the line, option (D).

Answer: (D)

6. Medium · Desmos

The table shows the age x, in years, and the efficiency loss y, in percent, for a sample of rooftop solar panels.

x	1	2	3	4	5
y	3.5	6.0	8.6	11.0	13.4

Which of the following equations could be the line of best fit for the data?

(A) $y = -2.5x + 1$

(B) $y = 2.5x - 1$

(C) $y = 2.5x + 1$ ✓

(D) $y = x + 2.5$

1. Open a table and enter $x_1 = 1, 2, 3, 4, 5$ and $y_1 = 3.5, 6.0, 8.6, 11.0, 13.4$.
2. Type `y_1~mx_1+b` : Desmos returns $m = 2.48$ and $b = 1.06$ (about 2.5 and 1).
3. Test option (C) at $x = 5$: Type `2.5*5+1` : it returns 13.5 against the data value 13.4.
4. The others miss badly at $x = 5$: Type `-2.5*5+1` : it returns -11.5, Type `5+2.5` : it returns 7.5, Type `2.5*5-1` : it returns 11.5.

Answer: (C)

7. Hard · Desmos

In which of the following tables is the relationship between the values of x and their corresponding y-values nonlinear?

(A)

x	1	2	3	4
y	5	10	15	20

(B)

x	1	2	3	4
y	3	6	12	24

✓

(C)

x	1	2	3	4
y	5	9	13	17

(D)

x	1	2	3	4
y	2	4	6	8

1. Linear means constant change. Enter each table and run `y_1~mx_1+b` . For the first three tables $r^2 = 1, 1, 1$ (perfect lines, with slopes 4, 2, 5).
2. For the fourth table ($y_1 = 3, 6, 12, 24$) the fit is not perfect: Desmos returns $r^2 = 0.92 < 1$.
3. Differences confirm it: Type `6-3` : it returns 3, Type `12-6` : it returns 6, Type `24-12` : it returns 12 are not equal (each y-value doubles).
4. The nonlinear relationship is the fourth table, option (B).

Answer: (B)

8. HARD · NO CALCULATOR NEEDED

A line of best fit models the relationship between x, the number of years since 2018, and y, the average monthly electricity subsidy, in dollars, provided to households in an energy-efficiency pilot program: $y = -18.6x + 340$. Which of the following is the best interpretation of the slope of this line of best fit in this context?

(A) The estimated subsidy decreases by \$340 each year.

(B) The estimated initial subsidy was \$18.6.

(C) The estimated subsidy increases by \$18.6 each year.

(D) The estimated subsidy decreases by \$18.6 each year. ✓

1. Type the model `y=-18.6x+340` (here x is years since 2018).
2. Evaluate at $x = 0$ and $x = 1$: Type `-18.6*0+340`: it returns 340, Type `-18.6*1+340`: it returns 321.4.
3. The change per year is Type `(-18.6*1+340)-(-18.6*0+340)`: it returns -18.6 dollars, a decrease. (Verbal reasoning step: the number 340 is the starting value, not a rate.) That is option (D).

Answer: (D)

9. HARD · DESMOS

Sam deposits \$800 into Account A, which earns 5% interest compounded annually. Sam also deposits \$1,500 into Account B, whose balance increases by a fixed \$40 each year. Based on these growth patterns, which of the following best describes the long-term behavior of the balances of Account A and Account B?

(A) The balance of Account A starts lower than the balance of Account B but eventually exceeds it. ✓

(B) The balance of Account A starts higher than the balance of Account B but eventually falls below it.

(C) The balance of Account A is always greater than the balance of Account B.

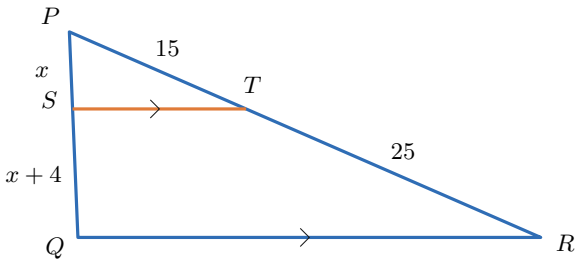
(D) The balance of Account A is always less than the balance of Account B.

1. Type `y=800*1.05^x` and `y=1500+40*x`.
2. Zoom out and tap the intersection with $x > 0$: (22.53, 2401.04) (about 22.5 years). The other intersection has a negative x, so ignore it.
3. At the start: Type `800*1.05^0`: it returns 800, Type `1500+40*0`: it returns 1500, so A begins lower.
4. Far out: Type `800*1.05^30`: it returns 3457.5539, Type `1500+40*30`: it returns 2700, so A has overtaken B. That is option (A).

Answer: (A)

10. HARD · DESMOS

In triangle PQR, segment ST is drawn parallel to side QR, where S lies on PQ and T lies on PR. If $PS = x$, $SQ = x + 4$, $PT = 15$, and $TR = 25$, what is the value of x?



1. $ST \parallel QR$ gives $\dfrac{x}{x+4} = \dfrac{15}{25}$. Type `y=x/(x+4)` and `y=15/25`. Tap the intersection: (6, 0.6).

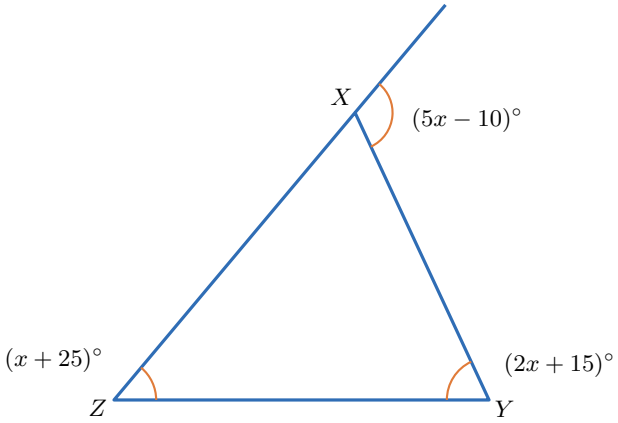
2. The first coordinate is $x = 6$.

3. Check: Type `(6)/((6)+4)` : it returns 0.6, Type `15/25` : it returns 0.6, equal.

Answer: 6

11. HARD · DESMOS

In triangle XYZ, the exterior angle at vertex X measures $(5x - 10)°$. This exterior angle equals the sum of the two non-adjacent (remote) interior angles, which measure $(2x + 15)°$ at vertex Y and $(x + 25)°$ at vertex Z. What is the measure of angle Y, in degrees?



1. Exterior angle = sum of the remote interior angles. Type `y=5x-10` and `y=(2x+15)+(x+25)` . Tap the intersection: (25, 115). Its first coordinate is the unknown $x = 25$.

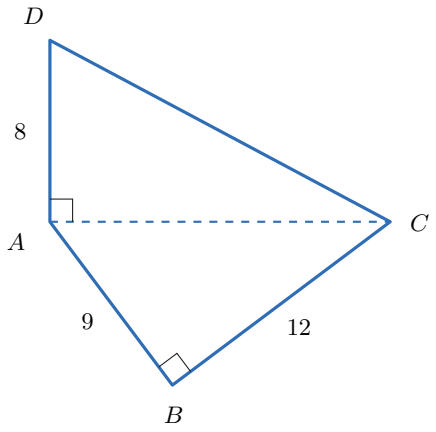
2. Angle $Y = 2x + 15$: Type `2(25)+15` : it returns 65 degrees.

3. Check the exterior angle: Type `5(25)-10` : it returns 115, Type `(2(25)+15)+((25)+25)` : it returns 115.

Answer: 65

12. HARD · DESMOS

In the figure, quadrilateral $ABCD$ is formed by two right triangles, ABC and ACD, sharing diagonal AC. Triangle ABC has a right angle at B with $AB = 9$ and $BC = 12$. Triangle ACD has a right angle at A with $AD = 8$. What is the area of quadrilateral $ABCD$?



(A) 108

(B) 114 ✓

(C) 90

(D) 96

1. The diagonal of the right triangle ABC: Type `sqrt(9^2+12^2)` : it returns 15.

2. Area of ABC: Type `1/2*9*12` : it returns 54.

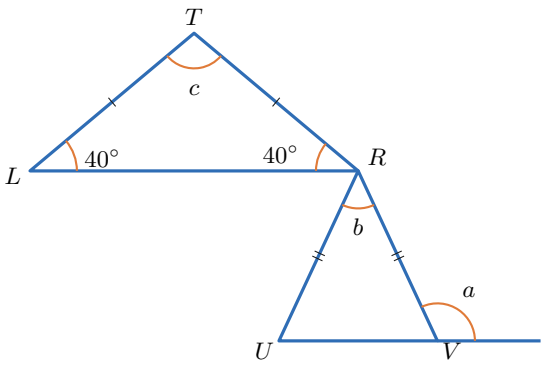
3. Area of ACD (legs $AC = 15$ and $AD = 8$): Type `1/2*15*8` : it returns 60.

4. Total: Type `1/2*9*12+1/2*15*8` : it returns 114, option (B).

Answer: (B)

13. HARD · DESMOS

In the figure shown, two isosceles triangles share the vertex R. In the top triangle, the vertex angle is c and each of its two equal base angles measures $40°$. In the bottom triangle, the vertex angle b is at R and equals half of c. Angle a is formed by extending the base of the bottom triangle, so a is supplementary to one of the bottom triangle's two equal base angles. What is the value of a, in degrees?



1. Top vertex angle: Type `180-40-40`: it returns 100 degrees.
2. Bottom vertex angle b: Type `100/2`: it returns 50 degrees.
3. Each bottom base angle: Type `(180-50)/2`: it returns 65 degrees.
4. Angle a is supplementary to a base angle: Type `180-65`: it returns 115.

Answer: 115

14. HARD · DESMOS

In triangle ABC, the exterior angle at vertex C measures $130°$. If the measure of angle A is $55°$, what is the measure of angle B?

1. Interior angle at C: Type `180-130`: it returns 50.
2. Angle B from the exterior angle theorem: Type `130-55`: it returns 75.
3. Check the angle sum: Type `55+75+50`: it returns 180.

Answer: 75

15. HARD · DESMOS

Lines l and m are parallel and are cut by transversal t. Which of the following pairs of measures could NOT represent the measures of two same-side interior angles formed by l, m, and t?

(A) 95° and 85°

(B) 112° and 68°

(C) 95° and 84° ✓

(D) 100° and 80°

1. Same-side interior angles must add to 180°. First two pairs: Type `100+80` : it returns 180, Type `95+85` : it returns 180.

2. Third and fourth pairs: Type `112+68` : it returns 180, Type `95+84` : it returns 179.

3. The pair summing to 179 cannot be same-side interior angles, option (C).

Answer: (C)

16. HARD · DESMOS

In triangle ABC, the exterior angle at vertex C measures $(7x-12)°$. This exterior angle equals the sum of the two non-adjacent interior angles, which measure $(3x+8)°$ and $(2x+16)°$. What is the value of x?

1. Type `y=7x-12` and `y=(3x+8)+(2x+16)` . Tap the intersection: (18, 114). Its first coordinate is the unknown $x = 18$.

2. Exterior angle: Type `7(18)-12` : it returns 114 degrees.

3. Remote angles: Type `(3(18)+8)+(2(18)+16)` : it returns 114, the same.

Answer: 18

17. HARD · DESMOS

A pole that is 2.4 meters tall casts a shadow 3.2 meters long. At the same time, a nearby communications tower casts a shadow 52 meters long. Using similar triangles, which of the following is closest to the height, in meters, of the tower?

(A) 39.0 ✓

(B) 41.6

(C) 34.7

(D) 36.4

1. Height per meter of shadow from the pole: Type `2.4/3.2` : it returns 0.75.

2. Tower height: Type `2.4/3.2*52` : it returns 39 meters.

3. (Flipping the ratio would give Type `3.2/2.4*52` : it returns 69.33, not an option.) That is option (A).

Answer: (A)

18. HARD · DESMOS

In isosceles triangle ABC, $AB = AC$, and the measure of angle A is $30°$ more than the measure of angle B. What is the measure, in degrees, of angle C?

1. Let each base angle be x (since $AB = AC$) and $A = x + 30$. Type `y=(x+30)+x+x` and `y=180`. Tap the intersection: (50, 180). Its first coordinate is the unknown $x = 50$.
2. Angle $C = x = 50$; angle A: Type `(50)+30`: it returns 80.
3. Check the sum: Type `80+50+50`: it returns 180.

Answer: 50

19. HARD · DESMOS

In triangle ABC, points M and N are the midpoints of sides AB and AC, respectively. If $MN = 3x - 4$ and $BC = 5x + 2$, what is the length of BC?

(A) 56 (B) 42

(C) 47 **(D) 52 ✓**

1. Midsegment: $MN = \frac{1}{2}BC$. Type `y=3x-4` and `y=(5x+2)/2`. Tap the intersection: (10, 26). Its first coordinate is the unknown $x = 10$.
2. BC: Type `5(10)+2`: it returns 52.
3. Check MN: Type `3(10)-4`: it returns 26, half of BC. That is option (D).

Answer: (D)

20. HARD · DESMOS

In $\triangle XYZ$, the side lengths are a, b, and c, where $4 < a < 6$ and $9 < b < 11$. Which of the following represents all possible values of c?

(A) $5 < c < 17$ (B) $3 < c < 15$

(C) $3 < c < 17$ ✓ (D) $5 < c < 15$

1. Triangle inequality: $|a - b| < c < a + b$. Smallest $|a - b|$ is near Type `abs(6-9)`: it returns 3.
2. Largest $a + b$ is near Type `6+11`: it returns 17.
3. Test values just inside the ranges: Type `abs(5.999-9.001)`: it returns 3.002, Type `5.999+10.999`: it returns 16.998 (never reaching 3 or 17). So $3 < c < 17$, option (C).

Answer: (C)

21. HARD · DESMOS

In triangle ABC, sides AC and AB are congruent. Points D, C, and B are collinear, with C between D and B. If the measure of angle DCA is 112°, what is the measure, in degrees, of angle A?

1. Angle ACB (supplementary to DCA): Type `180-112` : it returns 68.

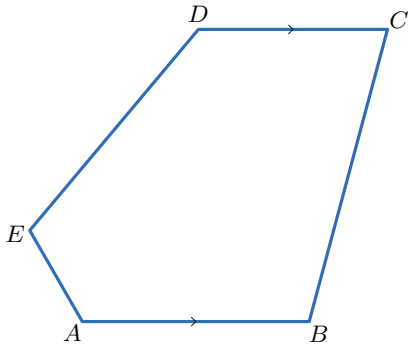
2. Angle A (isosceles, $B = ACB$): Type `180-68-68` : it returns 44.

3. Check the sum: Type `44+68+68` : it returns 180.

Answer: 44

22. HARD · DESMOS

In convex pentagon $ABCDE$ shown (not drawn to scale), side $\overline{AB}$ is parallel to side $\overline{DC}$. The measures of four interior angles are $\angle A = (6x)°$, $\angle B = (5x + 5)°$, $\angle C = (3x + 15)°$ and $\angle D = (7x - 10)°$. What is the measure, in degrees, of $\angle E$?



(A) 130

(B) 110 ✓

(C) 70

(D) 105

1. Same-side interior angles B and C add to 180. Type `y=(5x+5)+(3x+15)` and `y=180`. Tap the intersection: (20, 180). Its first coordinate is the unknown $x = 20$.

2. Angles: Type `6(20)` : it returns 120, Type `5(20)+5` : it returns 105, Type `3(20)+15` : it returns 75, Type `7(20)-10` : it returns 130.

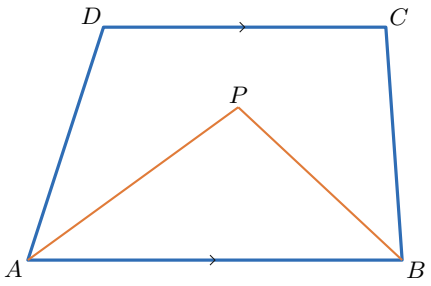
3. A pentagon's angles total Type `(5-2)*180` : it returns 540.

4. E: Type `540-(120+105+75+130)` : it returns 110, option (B).

Answer: (B)

23. HARD · DESMOS

In trapezoid $ABCD$ shown (not drawn to scale), $\overline{AB} \parallel \overline{DC}$, $\angle D = 108°$, and $\angle C = 94°$. The bisector of $\angle DAB$ and the bisector of $\angle ABC$ intersect at point P inside the trapezoid. What is the measure, in degrees, of the obtuse angle formed at P by these two bisectors?



(A) 79

(B) 22

(C) 108

(D) 101 ✓

1. $AB \parallel DC$: Type `180-108` : it returns 72, Type `180-94` : it returns 86 for A and B.
2. Half-angles: Type `72/2` : it returns 36, Type `86/2` : it returns 43.
3. Angle APB: Type `180-72/2-86/2` : it returns 101 (obtuse).
4. The obtuse angle is 101°, option (D). (The other angle at P is Type `180-101` : it returns 79.)

Answer: (D)

24. HARD · DESMOS

Two lines intersect at point O, and the smallest of the four angles they form measures 50°. A ray from O bisects each of the four angles. Together with the two lines, this gives eight rays from O, which divide the full turn into eight small angles. A *span* is an angle formed by 1, 2 or 3 consecutive small angles. Which of the following is NOT the measure of any span?

(A) 140° ✓

(B) 115°

(C) 130°

(D) 155°

1. Angles are 50 and 130; bisected: Type `50/2` : it returns 25, Type `(180-50)/2` : it returns 65.
2. The eight small angles add to a full turn: Type `total([25,25,65,65,25,25,65,65])` : it returns 360 (a full turn).
3. Spans of two or three: Type `25+25` : it returns 50, Type `25+65` : it returns 90, Type `65+65` : it returns 130, Type `25+25+65` : it returns 115, Type `25+65+65` : it returns 155.
4. The values 130, 155 and 115 occur; 140 does not. Option (A).

Answer: (A)

25. HARD · DESMOS

In equilateral triangle ABC, point D lies on side $\overline{BC}$ such that $BD = \frac{1}{4}BC$. What is the value of $\tan\angle DAC$?

(A) $\frac{3\sqrt{3}}{7}$ **(B) $\frac{3\sqrt{3}}{5}$ ✓**

(C) $\frac{\sqrt{3}}{7}$ (D) $\frac{5\sqrt{3}}{9}$

1. Coordinates $A = (2, 2\sqrt{3})$, $C = (4, 0)$, $D = (1, 0)$. Cross product of $\vec{AC} = (2, -2\sqrt{3})$ and $\vec{AD} = (-1, -2\sqrt{3})$: Type `abs(2*(-2*sqrt(3))-(-2*sqrt(3))*(-1))` : it returns 10.3923.
2. Dot product: Type `2*(-1)+(-2*sqrt(3))*(-2*sqrt(3))` : it returns 10.
3. Tangent: Type `abs(2*(-2*sqrt(3))-(-2*sqrt(3))*(-1))/(2*(-1)+(-2*sqrt(3))*(-2*sqrt(3)))` : it returns 1.0392.
4. Match to the options: Type `3*sqrt(3)/5` : it returns 1.0392 (option (B)); the others give Type `5*sqrt(3)/9` : it returns 0.9623, Type `3*sqrt(3)/7` : it returns 0.7423, Type `sqrt(3)/7` : it returns 0.2474.

Answer: (B)

26. HARD · DESMOS

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE}$ is parallel to $\overline{BC}$. The measure of $\angle ADE$ is $(4x - 10)°$, the measure of $\angle ABC$ is $(2x + 30)°$, and the measure of $\angle ACB$ is $(x + 25)°$. What is the measure, in degrees, of $\angle BAC$?

(A) 115 (B) 45

(C) 65 ✓ (D) 70

1. Corresponding angles: $\angle ADE = \angle ABC$. Type `y=4x-10` and `y=2x+30` . Tap the intersection: (20, 70). Its first coordinate is the unknown $x = 20$.
2. Angles B and C: Type `2(20)+30` : it returns 70, Type `(20)+25` : it returns 45.
3. $\angle BAC$: Type `180-70-45` : it returns 65, option (C).

Answer: (C)

27. HARD · DESMOS

Points A, B, C and D are consecutive vertices of a regular polygon with n sides, and $\overline{AC}$ is a diagonal. The measure of $\angle BAC$ is $(x + 3)°$ and the measure of $\angle ACD$ is $(10x + 15)°$, where x is a constant. What is the value of n?

(A) 12 ✓ (B) 15

(C) 24 (D) 8

1. With $x + 3 = \frac{180}{n}$ and $10x + 15 = 180 - \frac{540}{n}$, eliminating x gives $\frac{1800}{n} - 15 = 180 - \frac{540}{n}$. Type `y=1800/x-15` and `y=180-540/x` . Tap the intersection: (12, 135).
2. So $n = 12$.
3. Check the angles: Type `180/12` : it returns 15, Type `180-540/12` : it returns 135, Type `10*(180/12-3)+15` : it returns 135 (the last two agree). Option (A).

Answer: (A)

28. HARD · DESMOS

A convex polygon has n sides. Its interior angles, listed from smallest to largest, form an arithmetic sequence with smallest angle $120°$ and common difference $5°$. How many sides does the polygon have?

(A) 16 (B) 18

(C) 7 **(D) 9 ✓**

1. Sum of the angles: $\frac{n}{2}(2\cdot120+(n-1)5)$; the polygon sum is $180(n-2)$. Type `y=(x/2)(2*120+(x-1)*5)` and `y=180(x-2)`.

2. Tap both intersections: (9, 1260) and (16, 2520), so $n = 9$ or $n = 16$.

3. Largest angle for each: Type `120+5*(16-1)`: it returns 195, Type `120+5*(9-1)`: it returns 160.

4. Convex needs every angle under 180, so reject 16 and keep $n = 9$, option (D).

Answer: (D)

29. HARD · DESMOS

Two lines intersect and form four angles. The acute angles each measure $(2k + 4)°$, where k is a constant. Each obtuse angle measures 4 degrees more than 3 times the measure of an acute angle. Which of the following CANNOT be the sum, in degrees, of the measures of two of the four angles?

(A) 88 (B) 180

(C) 224 ✓ (D) 272

1. Type `y=(2x+4)+(3(2x+4)+4)` and `y=180`. Tap the intersection: (20, 180). Its first coordinate is the unknown $x = 20$.

2. Acute and obtuse angles: Type `2(20)+4`: it returns 44, Type `3*44+4`: it returns 136.

3. Sums of two angles: Type `44+44`: it returns 88, Type `44+136`: it returns 180, Type `136+136`: it returns 272.

4. 224 is not among them, so it cannot be a sum, option (C).

Answer: (C)

30. HARD · DESMOS

In triangle ABC, the measure of angle A is $(3x + 11)°$, the measure of angle B is $(2x + 4)°$, and the measure of angle C is $30°$. What is the value of

$$(\sin A\cos B + \cos A\sin B)(\cos A\cos B - \sin A\sin B)\,?$$

(A) $\frac{1}{4}$ **(B) $-\frac{\sqrt{3}}{4}$ ✓**

(C) $\frac{\sqrt{3}}{4}$ (D) $-\frac{1}{4}$

1. Type `y=(3x+11)+(2x+4)+30` and `y=180`. Tap the intersection: (27, 180). Its first coordinate is the unknown $x = 27$.

2. Angles A and B: Type `3(27)+11`: it returns 92, Type `2(27)+4`: it returns 58.

3. The two factors: Type `sin(92 deg)*cos(58 deg)+cos(92 deg)*sin(58 deg)`: it returns 0.5, Type `cos(92 deg)*cos(58 deg)-sin(92 deg)*sin(58 deg)`: it returns -0.866, and their product is Type `0.5*(-sqrt(3)/2)`: it returns -0.433.

4. Compare with option (B): Type `-sqrt(3)/4`: it returns -0.433.

Answer: (B)

31. Hard · Desmos

In triangle ABC, $AB = 12$ and $AC = 18$. The bisector of angle A meets $\overline{BC}$ at D, and $BC = 20$. What is the length of the longer of the two segments BD and DC?

1. Angle bisector theorem: $BD : DC = 12 : 18 = 2 : 3$, so $BD = 2k$, $DC = 3k$ with $5k = 20$. Type `y=2x+3x` and `y=20`. Tap the intersection: $(4, 20)$. Its first coordinate is the unknown $x = 4$.
2. BD and DC: Type `2(4)` : it returns 8, Type `3(4)` : it returns 12.
3. The longer segment is 12.

Answer: 12

32. Hard · Desmos

Two sides of a triangle have lengths 7 and 12. The third side has an integer length. How many different integer lengths are possible for the third side?

1. Triangle inequality bounds: Type `12-7` : it returns 5, Type `12+7` : it returns 19, so $5 < t < 19$.
2. Count the integers 6 to 18: Type `18-6+1` : it returns 13.
3. Let Desmos count them: Type `length([6...18])` : it returns 13.

Answer: 13

33. Hard · Desmos

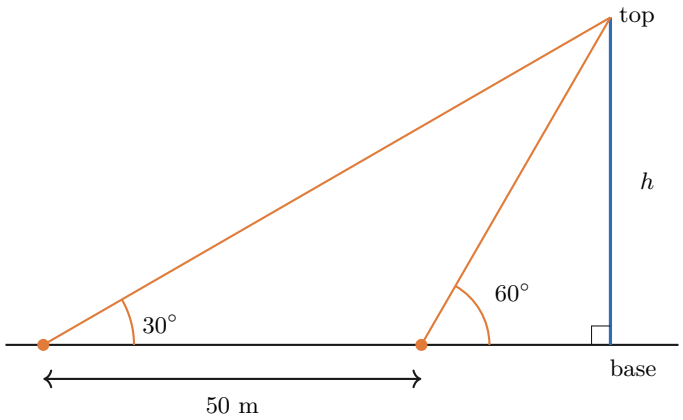
In triangle ABC, $AB = AC$ and angle A measures $40°$. Point D lies on $\overline{AC}$ such that $BD = BC$. What is the measure, in degrees, of angle ABD?

1. Base angles of ABC: Type `(180-40)/2` : it returns 70.
2. In triangle BCD ($BD = BC$): angle CBD is Type `180-2*70` : it returns 40.
3. $\angle ABD$: Type `70-40` : it returns 30.

Answer: 30

34. HARD · DESMOS

Engineers surveying a proposed tower measure the angle of elevation to the top of the tower from a point on the ground to be 30°. Standing 50 meters closer to the base of the tower, along the same straight line, the angle of elevation is 60°. What is the height of the tower, in meters, to the nearest tenth?

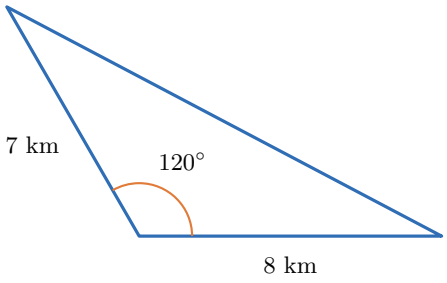


1. Let x be the distance to the base and y the height. Type the two sight lines. Type `y=x*tan(60 deg)` and `y=(x+50)*tan(30 deg)`. Tap the intersection: (25, 43.3013).
2. Read the tapped point: distance 25 and height 43.3013.
3. Check with an exact value: Type `25*sqrt(3)`: it returns 43.3013.
4. Round to the nearest tenth: Type `round(25*sqrt(3),1)`: it returns 43.3.

Answer: 43.3

35. HARD · DESMOS

A triangular plot of land has two sides of length 7 kilometers and 8 kilometers, with an included angle of 120° between them. What is the length, in kilometers, of the third side of the plot?



(A) 15 (B) 11

(C) 12 **(D) 13 ✓**

1. Law of Cosines: Type `7^2+8^2-2*7*8*cos(120 deg)`: it returns 169 for c^2.
2. Take the square root: Type `sqrt(7^2+8^2-2*7*8*cos(120 deg))`: it returns 13.
3. Option (D).

Answer: (D)

36. HARD · DESMOS

In right triangle JKL, the right angle is at L. If $\cos(J) = \frac{7}{25}$, what is the value of $\sin(J)$?



(A) $\frac{24}{25}$ ✓ (B) $\frac{25}{7}$

(C) $\frac{25}{24}$ (D) $\frac{7}{25}$

1. Missing leg: Type `sqrt(25^2-7^2)` : it returns 24.

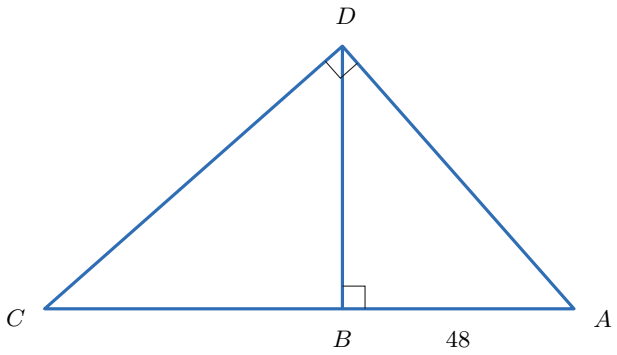
2. $\sin J$: Type `24/25` : it returns 0.96.

3. Identity check: Type `(24/25)^2+(7/25)^2` : it returns 1. Option (A).

Answer: (A)

37. HARD · DESMOS

In the figure shown, right triangle ACD has a right angle at D, and right triangle BCD, with B on segment AC, has a right angle at B. Both triangles share angle C. If $\cos C = 0.75$ and $AC = 48$, what is the length of BC?



1. $DC = \cos C \cdot AC$: Type `0.75*48` : it returns 36.

2. $BC = \cos C \cdot DC$: Type `0.75*36` : it returns 27.

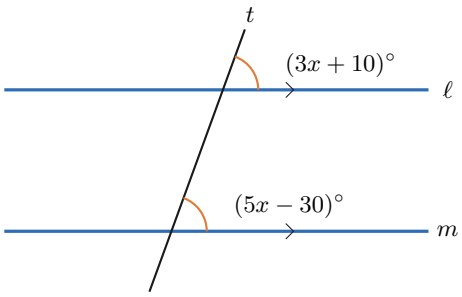
3. Check with the angle: Type `arccos(0.75)*180/pi` : it returns 41.4096 degrees for C, then Type `36*cos(41.4096 deg)` : it returns 27.

Answer: 27

PART B · SELF-ASSESSMENT — 24 PROBLEMS

38. MEDIUM · DESMOS

Line ℓ is parallel to line m and both are intersected by transversal t. One angle formed at the intersection with ℓ measures $(3x+10)°$, and its corresponding angle at the intersection with m measures $(5x-30)°$. What is the value of x?



(A) 15

(B) 20 ✓

(C) 40

(D) 10

1. Type `y=3x+10` and `y=5x-30`. Tap the intersection: (20, 70). Its first coordinate is the unknown $x = 20$.

2. Check both sides at $x = 20$: Type `3(20)+10`: it returns 70, Type `5(20)-30`: it returns 70.

3. Corresponding angles are equal, so $x = 20$ (option (B)).

Answer: (B)

39. MEDIUM · DESMOS

In triangle ABC, the measure of angle A is twice the measure of angle B, and the measure of angle C is 24° more than the measure of angle B. What is the measure of angle A, in degrees?

(A) 39

(B) 52

(C) 78 ✓

(D) 87

1. With $B = x$, $A = 2x$, $C = x + 24$ the angle sum is 180. Type `y=2x+x+(x+24)` and `y=180`. Tap the intersection: (39, 180). Its first coordinate is the unknown $x = 39$.

2. Angle $A = 2x$: Type `2(39)`: it returns 78 degrees.

3. Check the sum: Type `2(39)+(39)+((39)+24)`: it returns 180. That is option (C).

Answer: (C)

40. Medium · Desmos

A regular polygon has exterior angles that each measure 24°. How many sides does the polygon have?

(A) 15 ✓ (B) 18

(C) 24 (D) 12

1. The exterior angles sum to 360°, so each is $\frac{360}{n}$. Type `y=360/x` and `y=24`. Tap the intersection: (15, 24).

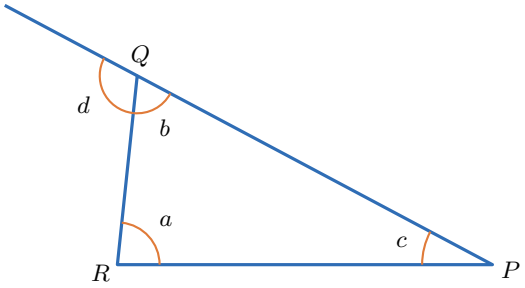
2. The number of sides is the first coordinate $n = 15$.

3. Check: Type `360/15`: it returns 24 degrees each. That is option (A).

Answer: (A)

41. Medium · Desmos

In the figure shown, a, b and c are the three interior angles of a triangle, with $a = 3c$. Side PQ of the triangle is extended through Q, and the angle d that this extension makes with the side at Q measures 112°, so b and d together form a straight line. What is the value of $a + b$?



(A) 158 (B) 140

(C) 146 **(D) 152 ✓**

1. $b = 180 - 112$: Type `180-112`: it returns 68.

2. With $c = x$, $a = 3x$: Type `y=3x+x+68` and `y=180`. Tap the intersection: (28, 180). Its first coordinate is the unknown $x = 28$.

3. So $c = 28$, $a = 3c$: Type `3(28)`: it returns 84.

4. $a + b$: Type `3(28)+68`: it returns 152, option (D).

Answer: (D)

42. Medium · Desmos

What is the positive difference, in degrees, between the measure of an exterior angle of a regular octagon (8 sides) and the measure of an exterior angle of a regular decagon (10 sides)?

1. Octagon exterior angle: Type `360/8`: it returns 45.

2. Decagon exterior angle: Type `360/10`: it returns 36.

3. Positive difference: Type `360/8-360/10`: it returns 9.

Answer: 9

43. Medium · Desmos

At a certain time of day, a 6-meter flagpole casts a shadow that is 4 meters long. At the same time, a nearby building casts a shadow that is 30 meters long. What is the height, in meters, of the building?

1. Similar triangles: height over shadow is constant. Type `6/4` : it returns 1.5 for the flagpole.

2. Building: Type `6/4*30` : it returns 45 meters.

3. Check the ratio: Type `45/30` : it returns 1.5, equal to 6/4.

Answer: 45

44. Medium · No calculator needed

Triangle DEF is similar to triangle GHI, where D, E, and F correspond to G, H, and I, respectively. The measure of angle D is 48°, and the measure of angle E is 63°. What is the measure of angle H?

(A) 117°

(B) 48°

(C) 63° ✓

(D) 69°

1. Similar triangles have equal corresponding angles, so $H = E = 63°$ (option (C)).

2. Angle F: Type `180-48-63` : it returns 69.

3. Angle sum check for GHI: Type `48+63+69` : it returns 180.

Answer: (C)

45. Medium · Desmos

In a triangle, the measures of the three interior angles are x, $2x$, and $(3x - 30)$ degrees. What is the value of x?

(A) 32

(B) 35 ✓

(C) 40

(D) 30

1. Type `y=x+2x+(3x-30)` and `y=180` . Tap the intersection: (35, 180). Its first coordinate is the unknown $x = 35$.

2. Largest angle: Type `3(35)-30` : it returns 75 degrees.

3. Check the sum: Type `(35)+2(35)+(3(35)-30)` : it returns 180. That is option (B).

Answer: (B)

46. Medium · Desmos

Line ℓ is parallel to line m, and a transversal crosses both lines. One angle formed at line ℓ measures $(4x+5)°$, and the corresponding angle formed at line m measures $(6x-21)°$. What is the value of x?

(A) 15

(B) 18

(C) 11

(D) 13 ✓

1. Type `y=4x+5` and `y=6x-21`. Tap the intersection: (13, 57). Its first coordinate is the unknown $x = 13$.

2. Check both angles: Type `4(13)+5`: it returns 57, Type `6(13)-21`: it returns 57.

3. So $x = 13$, option (D).

Answer: (D)

47. Medium · Desmos

Two lines intersect, forming four angles. One angle measures $(5x + 10)°$ and the angle vertical to it measures $(3x + 30)°$. What is the value of x?

(A) 10 ✓

(B) 12

(C) 15

(D) 8

1. Type `y=5x+10` and `y=3x+30`. Tap the intersection: (10, 60). Its first coordinate is the unknown $x = 10$.

2. Check: Type `5(10)+10`: it returns 60, Type `3(10)+30`: it returns 60.

3. So $x = 10$, option (A).

Answer: (A)

48. Medium · Desmos

A triangle has side lengths of 9, 14, and x, where x is an integer. What is the greatest possible value of x?

(A) 21

(B) 22 ✓

(C) 23

(D) 24

1. Triangle inequality: $x < 9 + 14$. Type `9+14`: it returns 23.

2. The greatest integer below 23: Type `23-1`: it returns 22.

3. Check the other inequalities for $x = 22$: Type `9+22`: it returns 31, Type `14+9`: it returns 23 (need $31 > 14$ and $23 > 22$). That is option (B).

Answer: (B)

49. Medium · Desmos

Triangles ABC and DEF are similar. Side AB of triangle ABC corresponds to side DE of triangle DEF. If the length of $\overline{AB}$ is 12 units and the length of $\overline{DE}$ is 8 units, then the area of triangle DEF is equal to

(A) $\frac{16}{9}$ (area of triangle ABC)

(B) $\frac{2}{3}$ (area of triangle ABC)

(C) $\frac{4}{9}$ (area of triangle ABC) ✓

(D) $\frac{9}{4}$ (area of triangle ABC)

1. Scale factor DEF to ABC sides: Type `8/12` : it returns 0.6667.
2. Area ratio is the square: Type `(8/12)^2` : it returns 0.4444.
3. Compare with the option $\frac{4}{9}$: Type `4/9` : it returns 0.4444, the same. Option (C).

Answer: (C)

50. Medium · Desmos

Two parallel lines are cut by a transversal. A pair of alternate interior angles measure $(6x - 8)°$ and $(4x + 12)°$. What is the measure, in degrees, of each of these angles?

1. Alternate interior angles are equal. Type `y=6x-8` and `y=4x+12` . Tap the intersection: (10, 52). Its first coordinate is the unknown $x = 10$.
2. Each angle: Type `6(10)-8` : it returns 52 degrees.
3. Check: Type `4(10)+12` : it returns 52.

Answer: 52

51. Medium · Desmos

An exterior angle of a triangle measures $(8x-6)°$. The two remote interior angles measure $(3x+14)°$ and $(2x+19)°$. What is the measure, in degrees, of the exterior angle?

1. Exterior angle = sum of remote interior angles. Type `y=8x-6` and `y=(3x+14)+(2x+19)` . Tap the intersection: (13, 98). Its first coordinate is the unknown $x = 13$.
2. Exterior angle: Type `8(13)-6` : it returns 98 degrees.
3. Check the remote sum: Type `(3(13)+14)+(2(13)+19)` : it returns 98.

Answer: 98

52. Medium · Desmos

In an isosceles triangle, the two base angles each measure $(x+20)°$ and the vertex angle measures $(4x-10)°$. What is the measure, in degrees, of the vertex angle?

1. Type `y=(4x-10)+2(x+20)` and `y=180` . Tap the intersection: (25, 180). Its first coordinate is the unknown $x = 25$.
2. Vertex angle: Type `4(25)-10` : it returns 90 degrees.
3. Base angle and sum check: Type `(25)+20` : it returns 45, Type `90+45+45` : it returns 180.

Answer: 90

53. Medium · Desmos

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ such that $\overline{DE}$ is parallel to $\overline{BC}$. If $AD = 10$, $DB = 6$ and $AE = 15$, what is the length of $\overline{EC}$?

1. $AB = AD + DB$: Type `10+6`: it returns 16. With $DE \parallel BC$, $\dfrac{AD}{AB} = \dfrac{AE}{AC}$, so $AE = \frac{10}{16}AC$.

2. Type the ratio and $AE = 15$: Type `y=10/16*x` and `y=15`. Tap the intersection: (24, 15), so $AC = 24$.

3. $EC = AC - AE$: Type `24-15`: it returns 9.

Answer: 9

54. Medium · Desmos

Each interior angle of a regular polygon measures 140°. How many sides does the polygon have?

1. Each interior angle of a regular n-gon is $\dfrac{180(n-2)}{n}$. Type `y=180(x-2)/x` and `y=140`. Tap the intersection: (9, 140).

2. The number of sides is $n = 9$.

3. Check via exterior angles: Type `180-140`: it returns 40, Type `360/40`: it returns 9.

Answer: 9

55. Medium · Desmos

A trapezoid has parallel sides of length 10 and 24. A segment joins the midpoints of the two non-parallel sides. What is the length of this segment?

(A) 17 ✓ (B) 34

(C) 14 (D) 7

1. The midsegment is the average of the bases: Type `mean([10,24])`: it returns 17.

2. Check: Type `10+24`: it returns 34, Type `(10+24)/2`: it returns 17.

3. Option (A).

Answer: (A)

56. Medium · Desmos

In the xy-plane, a triangle has vertices $(0, 0)$, $(8, 0)$ and $(3, k)$, where $k > 0$. The area of the triangle is 20. What is the value of k?

1. Area $= \frac{1}{2}(8)(k)$ with the x-coordinate 3 not mattering. Type `y=1/2*8*x` and `y=20`. Tap the intersection: (5, 20). Its first coordinate is the unknown $x = 5$.

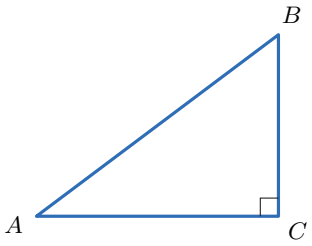
2. So $k = 5$.

3. Check the area: Type `1/2*8*5`: it returns 20.

Answer: 5

57. Medium · Desmos

In right triangle ABC, the right angle is at C. If $\sin(A) = \frac{3}{5}$, what is the value of $\cos(B)$?



(A) $\frac{4}{5}$

(B) $\frac{5}{3}$

(C) $\frac{3}{4}$

(D) $\frac{3}{5}$ ✓

1. Angle A in degrees: Type `arcsin(3/5)*180/pi` : it returns 36.8699.

2. Angle B (complementary): Type `90-36.8699` : it returns 53.1301.

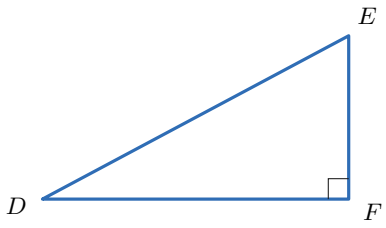
3. $\cos B$: Type `cos(53.1301 deg)` : it returns 0.6.

4. This equals Type `3/5` : it returns 0.6, so $\cos B = \sin A$. Option (D).

Answer: (D)

58. Medium · Desmos

In right triangle DEF, the right angle is at F. If $\tan(D) = \frac{8}{15}$, what is the value of $\tan(E)$?



(A) $\frac{17}{15}$

(B) $\frac{8}{15}$

(C) $\frac{15}{8}$ ✓

(D) $\frac{8}{17}$

1. Angle D: Type `arctan(8/15)*180/pi` : it returns 28.0725 degrees.

2. Angle $E = 90 - D$: Type `90-28.0725` : it returns 61.9275.

3. $\tan E$: Type `tan(61.9275 deg)` : it returns 1.875.

4. Compare with Type `15/8` : it returns 1.875. Option (C).

Answer: (C)

59. Medium · Desmos

If θ is an acute angle and $\sin(\theta) = \frac{5}{13}$, what is the value of $\tan(\theta)$?

(A) $\frac{12}{13}$

(B) $\frac{5}{12}$ ✓

(C) $\frac{12}{5}$

(D) $\frac{5}{13}$

1. Adjacent side: Type `sqrt(13^2-5^2)` : it returns 12.

2. $\tan\theta$: Type `5/12` : it returns 0.4167.

3. Direct check: Type `tan(arcsin(5/13))` : it returns 0.4167. Option (B).

Answer: (B)

60. Medium · No calculator needed

In a 30°-60°-90° right triangle, the side opposite the 30° angle has length 7. What is the length of the hypotenuse?

(A) 7

(B) $7\sqrt{2}$

(C) $7\sqrt{3}$

(D) 14 ✓

1. The hypotenuse is twice the short side: Type `2*7` : it returns 14.

2. Check with sine: Type `7/sin(30 deg)` : it returns 14.

3. Long leg check: Type `sqrt(14^2-7^2)` : it returns 12.1244 equals Type `7*sqrt(3)` : it returns 12.1244. Option (D).

Answer: (D)

61. Medium · No calculator needed

What is the value of $\cos(60°) + \sin(30°)$?

(A) 1 ✓

(B) $\sqrt{3}$

(C) $\frac{1}{2}$

(D) $\frac{\sqrt{3}}{2}$

1. Evaluate each term: Type `cos(60 deg)` : it returns 0.5, Type `sin(30 deg)` : it returns 0.5.

2. Add them: Type `cos(60 deg)+sin(30 deg)` : it returns 1.

3. Option (A).

Answer: (A)